Primary Servos (S.O.O. 6221) – Maintenance Practices

1. Description

A. Flight Control Primary Servos (SM3000)

The aileron (roll), elevator (pitch) and rudder (yaw) fail passive, primary servos, move the associated aircraft control surfaces in response to signals from the AIOP (Actuator Input Output Processor) with the MAU. Communication between the MAU and primary servos is over bidirectional CAN (Control Area Network) Buses, motor enable discrete and clutch enable discrete.

Each servo contains a DC motor, gearing and an electromechanical clutch. The electromechanical clutch is engaged when 28 volts is applied to the clutch supply. The clutch only engages or disengages the servo; torque limiting is done electronically and controlled through the Apex software. Each servo is mounted on and drives a capstan (SB3000), which actuates the associated aircraft control surfaces through servo bridle cables clamped to the control cables.

28 Volt DC power is protected by a 7.5 Amp AP SERVOS (P1) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

2. Removal/Installation

Refer to Figure 1 and Figure 2 for details.

A. Remove Primary Servos (Rudder, Elevator or Aileron)

Note The primary servos (rudder, elevator or aileron) can be replaced without removing the capstan and disturbing the cable rigging.

1) Ensure all aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to servos as follows:

a) Rudder and Elevator

1 Remove bulkhead panel at station 406 of rear baggage compartment.

2 Remove aft baggage extension floor panels.

b) Aileron

1 Remove Co-pilot's seat. (Refer to 25-10-00, Flight Compartment – General Data, Para 3.).

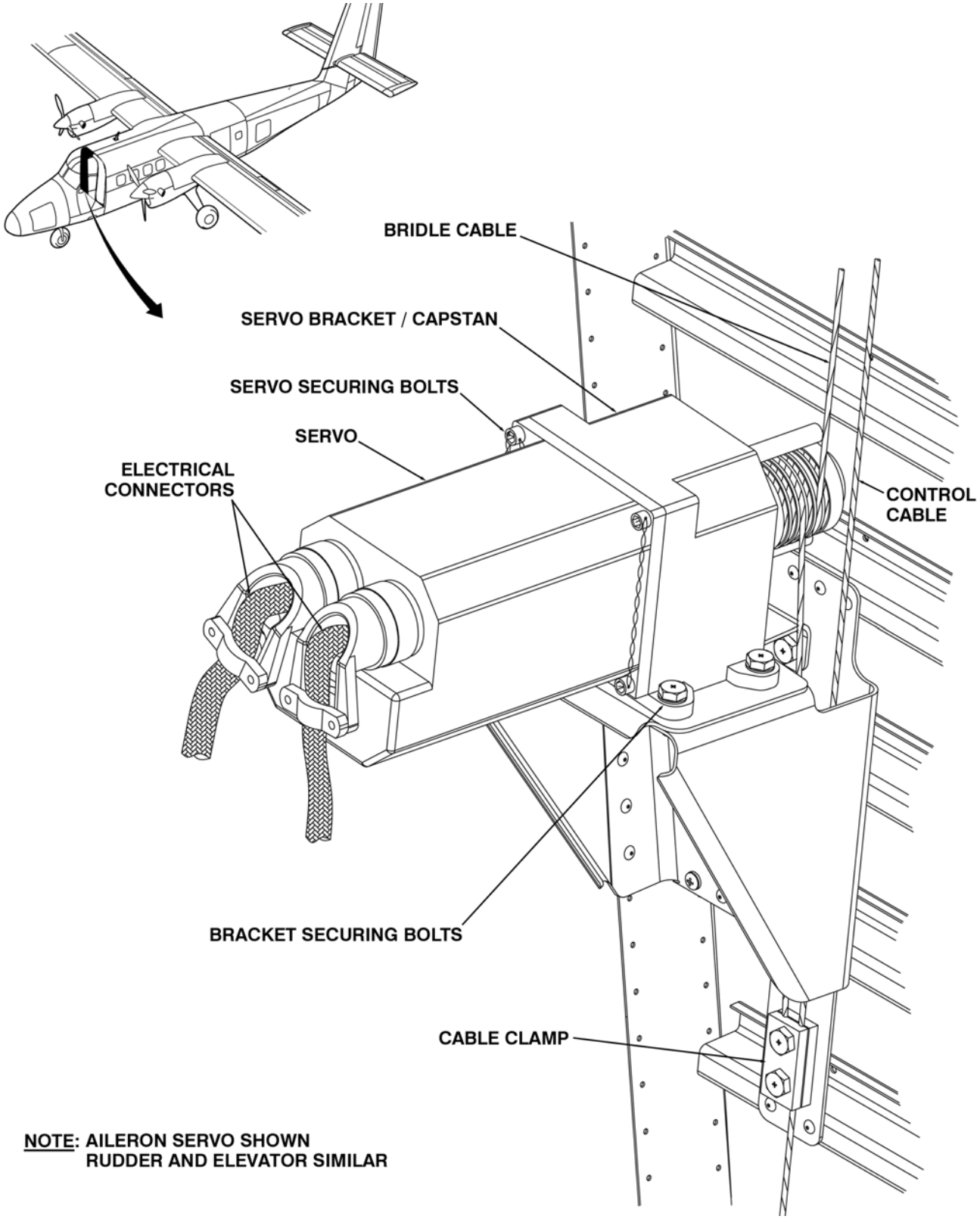
2 Remove aileron cable covers on Co-pilot side in flight compartment.

3) Disconnect electrical connectors from servos.

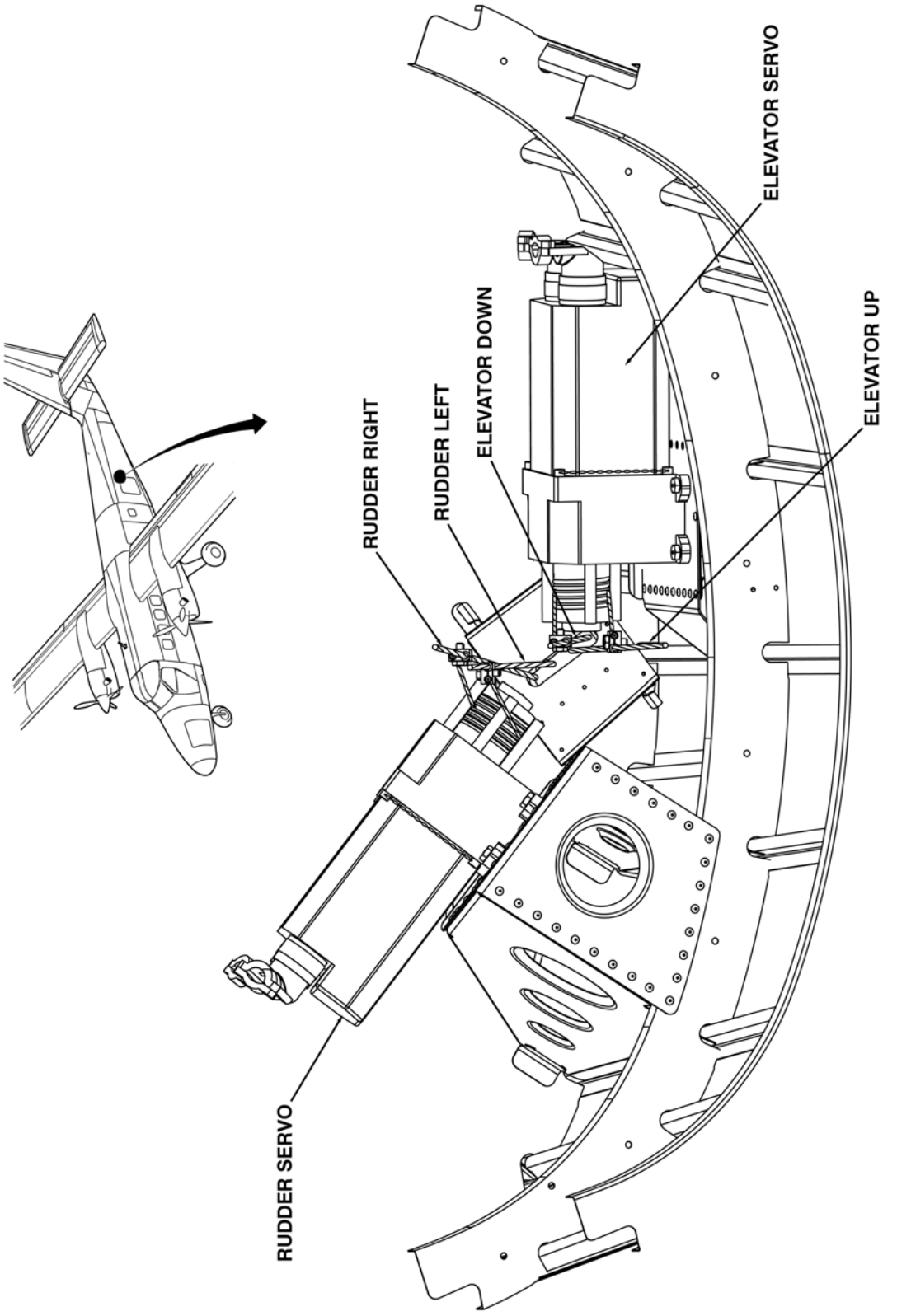
4) Remove lock wire from captive head screws securing the servo to the capstan.

5) Support servo, remove captive screws securing servo to capstan and withdraw servo, taking care not to damage splines on the servo drive shaft or capstan.

6) Remove and discard servo O-ring.



Primary Servo Installation
Figure 1



Completed Rudder and Elevator Servo Installation
Figure 2

Print Date: 2025-09-12

B. Install Primary Servos (Rudder, Elevator or Aileron)

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED.**

1) Ensure all aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Servo to airframe electrical bonding is maintained through direct contact between the servo and airframe mating surfaces. Prior to installation, ensure mating surfaces are clean and free of contaminants. Refer to procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data, as required.

3) Ensure servo capstan is correctly rigged. (Refer to 22-10-12, Primary Servo Capstan (S.O.O. 6221) – General Data, Figure 1, Figure 2 and Figure 3).

4) Install new servo O-ring, apply thin coat of Krytox 283 AD grease or equivalent to internal spline, O-ring on servo body and pilot diameter on servo capstan.

5) Engage internal spline on servo drive shaft with spline on servo capstan shaft and secure with captive screws. Torque screws to 29±3 in-lbs and lockwire.

6) Connect electrical connectors to servo.

7) Do bonding check. (Refer to Para 3.A).

8) Restore aircraft electrical power. (Refer to 24-00-00, Electrical Power – General Data).

9) Function test system. (Refer to 22-10-00, Automatic Flight Controls System (AFCS) (Mod 6/2303 and S.O.O. 6221) – General Data, Para 3.C. and Para 3.F.).

10) If rudder or elevator servo has been installed, install aft baggage extension floor panels and bulkhead panel at station 406 of rear baggage compartment. If aileron servo has been installed, install aileron cable covers in flight compartment and install Co-pilot's seat.

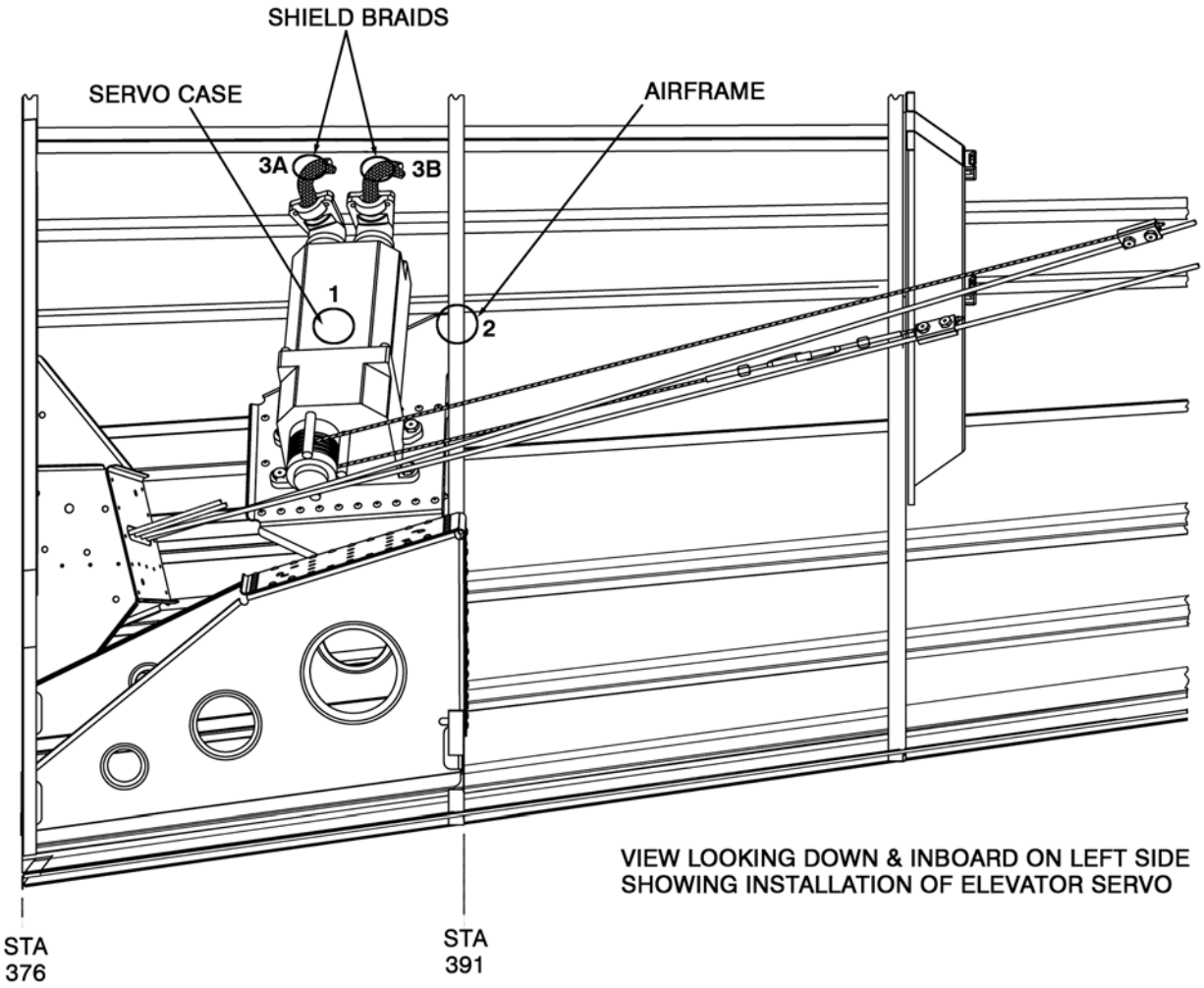
3. Adjustment/Test

A. Electrical Bonding

Refer to Figure 3 for Elevator Bonding Check.

Refer to Figure 4 for Rudder Bonding Check.

Refer to Figure 5 for Aileron Bonding Check.



CONDUCT THE FOLLOWING ELECTRICAL BONDING CHECKS:

SERVO CASE TO AIRFRAME (POINT 1 TO POINT 2) REFER TO NOTES 1 AND 2. IMPEDANCE BOND VALUE TO BE LESS THAN 2.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3.

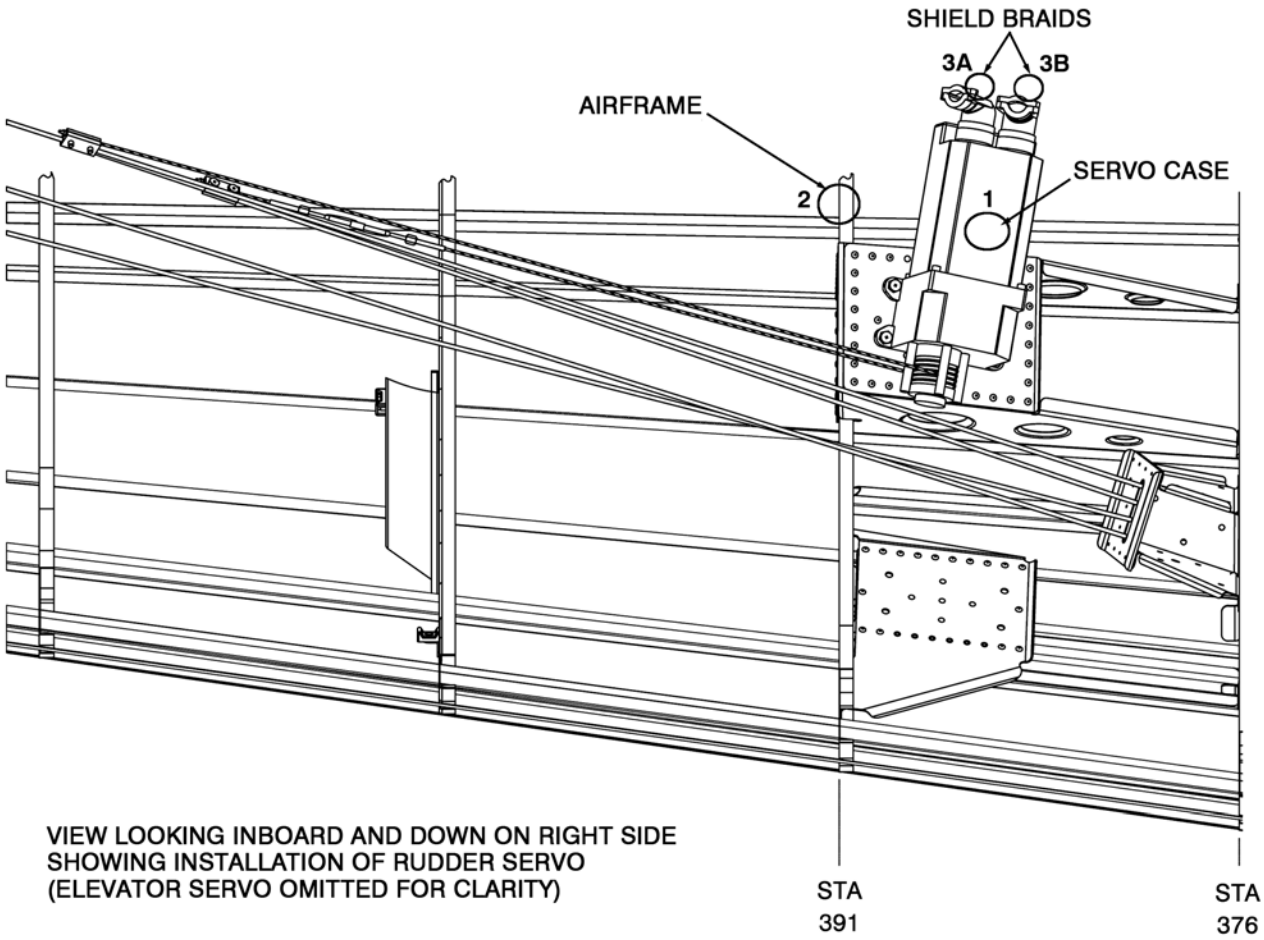
SHIELD BRAIDS TO SERVO CASE (POINT 3A TO POINT 1) REFER TO NOTE 1. IMPEDANCE BOND VALUE TO BE LESS THAN 7.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3. REPEAT FOR POINT 3B TO POINT 1.

NOTE 1: SERVO PROBE CONTACT POINT SHOULD BE SELECTED ON SERVO CASE. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT POINT. SERVO ASSEMBLY FASTENER MAY BE USED.

NOTE 2: AIRFRAME PROBE CONTACT POINT SHOULD BE SELECTED AS CLOSE AS POSSIBLE TO SERVO BRACKET, BUT NOT ON THE MOUNTING BRACKET. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT TO AIRFRAME GROUND.

NOTE 3: IF REQUIRED, THE MAXIMUM STATED IMPEDANCE VALUES MAY BE ADJUSTED PER THE FOLLOWING; AN ACCEPTABLE IMPEDANCE BOND VALUE (MAXIMUM MILLIOHMS) MAY BE DETERMINED BY MULTIPLYING THE NUMBER OF CONTACT BOND SURFACES BY 2.5 MILLIOHMS FOR EACH BOND SURFACE LOCATED BETWEEN THE PROBE CONTACT POINTS.

Elevator Bonding Check
Figure 3



VIEW LOOKING INBOARD AND DOWN ON RIGHT SIDE
SHOWING INSTALLATION OF RUDDER SERVO
(ELEVATOR SERVO OMITTED FOR CLARITY)

CONDUCT THE FOLLOWING ELECTRICAL BONDING CHECKS:

SERVO CASE TO AIRFRAME (POINT 1 TO POINT 2) REFER TO NOTES 1 AND 2. IMPEDANCE BOND VALUE TO BE LESS THAN 2.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3.

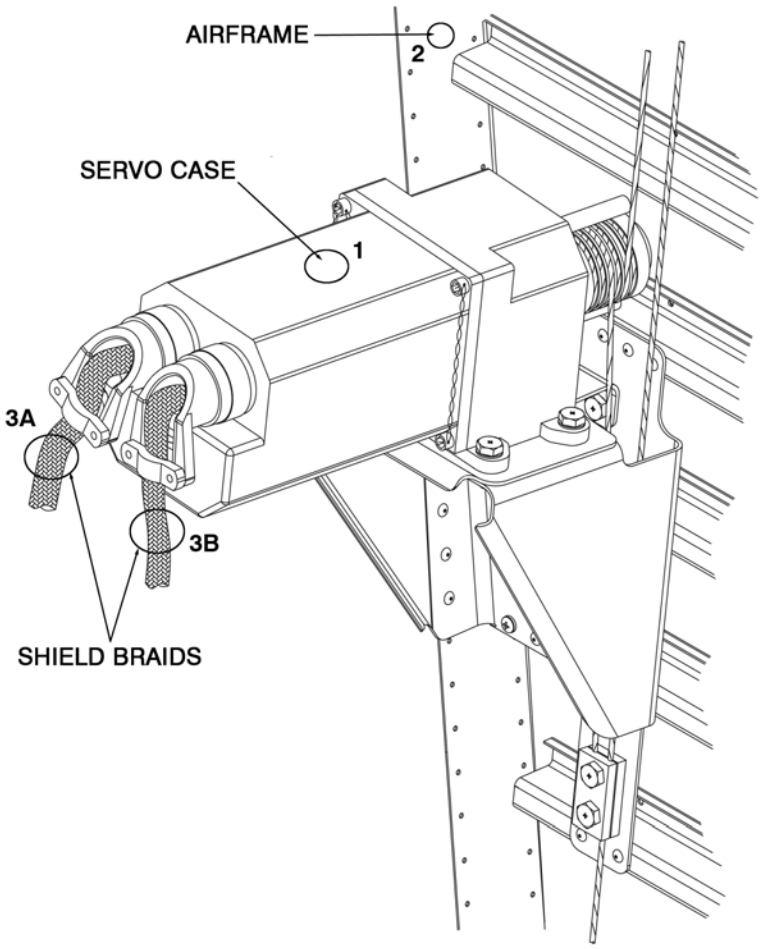
SHIELD BRAIDS TO SERVO CASE (POINT 3A TO POINT 1) REFER TO NOTE 1. IMPEDANCE BOND VALUE TO BE LESS THAN 7.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3. REPEAT FOR POINT 3B TO POINT 1.

NOTE 1: SERVO PROBE CONTACT POINT SHOULD BE SELECTED ON SERVO CASE. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT POINT. SERVO ASSEMBLY FASTENER MAY BE USED.

NOTE 2: AIRFRAME PROBE CONTACT POINT SHOULD BE SELECTED AS CLOSE AS POSSIBLE TO SERVO BRACKET, BUT NOT ON THE MOUNTING BRACKET. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT TO AIRFRAME GROUND.

NOTE 3: IF REQUIRED, THE MAXIMUM STATED IMPEDANCE VALUES MAY BE ADJUSTED PER THE FOLLOWING; AN ACCEPTABLE IMPEDANCE BOND VALUE (MAXIMUM MILLIOHMS) MAY BE DETERMINED BY MULTIPLYING THE NUMBER OF CONTACT BOND SURFACES BY 2.5 MILLIOHMS FOR EACH BOND SURFACE LOCATED BETWEEN THE PROBE CONTACT POINTS.

Rudder Bonding Check
Figure 4



CONDUCT THE FOLLOWING ELECTRICAL BONDING CHECKS:

SERVO CASE TO AIRFRAME (POINT 1 TO POINT 2) REFER TO NOTES 1 AND 2. IMPEDANCE BOND VALUE TO BE LESS THAN 2.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3.

SHIELD BRAIDS TO SERVO CASE (POINT 3A TO POINT 1) REFER TO NOTE 1. IMPEDANCE BOND VALUE TO BE LESS THAN 7.5 MILLIOHMS. IF NOT ACHIEVABLE, REFER TO NOTE 3. REPEAT FOR POINT 3B TO POINT 1.

NOTE 1: SERVO PROBE CONTACT POINT SHOULD BE SELECTED ON SERVO CASE. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT POINT. SERVO ASSEMBLY FASTENER MAY BE USED.

NOTE 2: AIRFRAME PROBE CONTACT POINT SHOULD BE SELECTED AS CLOSE AS POSSIBLE TO SERVO BRACKET, BUT NOT ON THE MOUNTING BRACKET. REMOVE NON-CONDUCTIVE COATING, IF REQUIRED, TO ACHIEVE AN ACCEPTABLE CONTACT TO AIRFRAME GROUND.

NOTE 3: IF REQUIRED, THE MAXIMUM STATED IMPEDANCE VALUES MAY BE ADJUSTED PER THE FOLLOWING; AN ACCEPTABLE IMPEDANCE BOND VALUE (MAXIMUM MILLIOHMS) MAY BE DETERMINED BY MULTIPLYING THE NUMBER OF CONTACT BOND SURFACES BY 2.5 MILLIOHMS FOR EACH BOND SURFACE LOCATED BETWEEN THE PROBE CONTACT POINTS.

Aileron Bonding Check
Figure 5

This Page Intentionally Left Blank